

Econometrics Preliminary Exam, August 2025

There are **FOUR** questions. Answer each part of each question. All questions are weighted equally. Within each question, each part will receive equal weight in grading. You have 20 minutes to read the exam and then four hours to complete the exam.

1 Probability and statistics

- (a) Consider random variables Z_1 , Z_2 , and X_1 , X_2 , where $Z_1, Z_2 \sim i.i.d. U[0, 1]$, $X_1 = 1(Z_1 \leq q)$, and $X_2 = 1(Z_2 \geq q)$; $0 < q < 1$. Let $Z = Z_1 + Z_2$. Let $X = X_1 + X_2$.
1. What distributions do random variables X_1 and X_2 follow? Derive $\mathbb{E}[X]$ and $\mathbb{V}[X]$.
 2. Find the support of X . Derive the probability mass function of X .
 3. Derive $\mathbb{E}[X_1|Z_1 = z]$ for all $0 < z < 1$. Derive $\mathbb{E}[Z_1|X_1 = x]$ for both $x = 0, 1$.
 4. Find the support of Z . Derive the probability density function of Z .
- (b) Suppose you observe a random sample $Y_1, \dots, Y_n$ drawn from a discrete distribution with probability mass function $\mathbb{P}[Y_i = -1] = p$, $\mathbb{P}[Y_i = 1] = 2p$, $\mathbb{P}[Y_i = 0] = 1 - 3p$.
5. What is the range of p ? Propose a moment estimator of p . Call your estimator $\hat{p}_{MOM}$.
 6. Show that $\sqrt{n}(\hat{p}_{MOM} - p) \xrightarrow{d} N(0, V)$ as $n \rightarrow \infty$. Derive the asymptotic variance V and provide an estimator for it. Show consistency of your proposed variance estimator.
 7. Derive the maximum likelihood estimator of p . Call your maximum likelihood estimator $\hat{p}_{MLE}$.
 8. Derive the bias and variance of both $\hat{p}_{MOM}$ and $\hat{p}_{MLE}$. Compare the two estimators in terms of bias and variance.

2 Linear models

Suppose that you observe an i.i.d. sample (Y_i, X_i) , $i = 1, \dots, n$, where Y_i and X_i are scalar random variables that have a joint normal distribution. Let $Y_i = X_i\beta + \varepsilon_i$, $\mathbb{E}(\varepsilon_i) = \mathbb{E}(X_i) = 0$, $\mathbb{E}(\varepsilon_i X_i) = 0$, $\mathbb{E}(\varepsilon_i^2) = \sigma^2$.

1. Is the OLS estimator $\hat{\beta}$ unbiased for β ? If yes, prove the result. If not, state any additional assumptions required.
2. Is this model conditionally homoscedastic under the stated assumptions? Explain why or why not (provide a counterexample if necessary). (Hint: Use properties of the normal distribution to examine the independence property.)
3. Is the OLS estimator $\hat{\beta}$ efficient under the maintained assumptions? If so, state the class of estimators in which it is efficient.
4. Propose an unbiased estimator for σ^2 . You do not need to prove unbiasedness for this question.
5. Derive the distribution of $\hat{\beta}$ conditional on $\{X_i\}_{i=1}^n$.
6. Propose a statistic for an exact test of the hypothesis $\beta = 0$ against its two-sided alternative. Make sure to only use feasible estimators in your proposal.
7. Specify critical values for the test in the previous part of the question with size $\alpha > 0$.
8. Explain how to use this test to construct a $1 - \alpha$ confidence set for β .

3 Nonlinear hypothesis testing and nonlinear models

- (a) Consider an OLS estimator $(\hat{\beta}_1, \hat{\beta}_2)$ of (β_1, β_2) that have asymptotic normal distribution with variance $V_{\hat{\beta}}$. Suppose that you have a consistent estimator $\hat{V}_{\hat{\beta}}$ of the matrix $V_{\hat{\beta}}$.
1. Find the asymptotic distribution of $\hat{\theta} = \hat{\beta}_1/\hat{\beta}_2$. State any additional assumptions required for your result.
 2. Propose a Wald test of hypothesis $\beta_1/\beta_2 = \theta_0$. Be explicit about the alternative hypothesis and the choice of the critical values.
 3. Would your answer to the previous question change if you reformulate the null hypothesis as $\beta_1 = \theta_0\beta_2$.
 4. Discuss the relative merits of the two alternative formulations of the null hypothesis.
- (b) Suppose we have i.i.d. observations $\{(Y_i, X_i) : i = 1, \dots, n\}$ with $Y_i \in \{0, 1\}$, $\dim(X_i) > 1$, and $\|X_i\| \leq 1$ a.s.¹ Assume $\mathbb{E}[Y | X] = \exp(X'\beta_0)/(1 + \exp(X'\beta_0))$ for some column vector β_0 . Suppose X satisfies the “no-multicollinearity” condition, i.e., $P(X'\beta_0 \neq X'\beta) > 0$ when $\beta \neq \beta_0$.

Consider the logistic regression estimator.

1. Prove the identification of β_0 . In the proof, clarify where you use the no-multicollinearity condition.
2. Prove that the logistic regression estimator is consistent. Make sure to state the sufficient conditions for the result.
3. Prove that the logistic regression estimator is asymptotically normal. Make sure to state the sufficient conditions for the result.
4. Propose an asymptotic variance estimator for the logistic regression estimator. You do not need to prove anything for this question.

¹These assumptions are stronger than those often imposed in the literature.

4 Panel data

- (a) Suppose we have a panel dataset $\{(Y_{it}, X_{it}) : i = 1, \dots, N, t = 1, \dots, T\}$ with $\dim(Y_{it}) = 1$, $\dim(X_{it}) > 1$, $E[Y_{it}^4] < \infty$ and $E[\|X_{it}\|^4] < \infty$. Suppose the observations are i.i.d. across i . All asymptotic arguments in this part of the question pertain to the case where $N \rightarrow \infty$ holding T fixed.

Consider the model $Y_{it} = X'_{it}\beta_0 + u_i + \varepsilon_{it}$. Assume $\mathbb{E}[X_{it}\varepsilon_{it}] = 0$ for every $s, t = 1, \dots, T$.

1. Write down the moment equation for the fixed effects estimator. You do not need to prove anything for this question.
 2. Provide a sufficient condition for the consistency of the fixed effects estimator. Prove that it is sufficient. Do not use the results in ARE/ECN 240B Reference Sheet.
 3. Write down the asymptotic variance of the fixed effects estimator. You do not need to prove anything for this question.
 4. In addition, assume $\mathbb{E}[u_i + \varepsilon_{it} \mid X_{i1}, \dots, X_{iT}] = 0$ for every $t = 1, \dots, T$. Explain why the (pooled) OLS estimator can be consistent. In your answer, derive the probability limit of the (pooled) OLS estimator.
- (b) We have clustered observations $\{(Y_{ig}, X_{ig}) : i = 1, \dots, N, g = 1, \dots, G\}$. We assume the clusters are equal-sized and that $\{(Y_{ig}, X_{ig}) : i = 1, \dots, N\}$ are i.i.d. across g .² We consider the asymptotic framework with $G \rightarrow \infty$ while N is fixed.

Consider the pooled OLS regression using this dataset.

1. Write down the model for the pooled OLS regression. Specify the parameter of interest.
2. Provide a sufficient condition for the consistency of the pooled regression estimator for the parameter of interest. Prove that it is sufficient.
3. Write down the asymptotic variance of the pooled OLS estimator. You do not need to prove anything for this question.
4. Explain how the asymptotic variance would be simplified when we impose the observations are independent within each cluster.

²These assumptions are stronger than those often imposed in the literature.

Probit	Logit
$\phi(u) = \frac{1}{\sqrt{2\pi}} \exp(-u^2/2)$	
$\Phi(u) = \int_{-\infty}^u \phi(t) dt$	$\Lambda(u) = (1 + \exp(-u))^{-1}$
$\frac{d}{du} \log \Phi(u) = \lambda(u)$ with $\lambda = \phi/\Phi$	$\frac{d}{du} \log \Lambda(u) = 1 - \Lambda(u)$
$\frac{d^2}{du^2} \log \Phi(u) = -\lambda(u)(u + \lambda(u)) < 0$	$\frac{d^2}{du^2} \log \Lambda(u) = -\Lambda(u)(1 - \Lambda(u)) < 0$
$\frac{d}{du} \phi(u) = -u\phi(u)$	
	$\frac{d}{du} \Lambda(u) = \Lambda(u)(1 - \Lambda(u))$
$\exists C > 0$	
$\sup_u \lambda(u) < C$	
$\sup_u \lambda(u)(u + \lambda(u)) < C$	
$ \log \Phi(u) \leq C(1 + u) u $	

Matrix Calculus: (i) $\frac{\partial}{\partial x}(a'x) = \frac{\partial}{\partial x}(x'a) = a$. (ii) $\frac{\partial}{\partial x}(x'A) = A$ and $\frac{\partial}{\partial x'}(Ax) = A$. (iii) $\frac{\partial}{\partial x}(x'Ax) = (A + A')x$. (iv) $\frac{\partial^2}{\partial x \partial x'}(x'Ax) = A + A'$.

Theorem 2.1. If there is a function $Q_0(\theta)$ such that

1. $Q_0(\theta)$ is uniquely maximized at θ_0 ;
2. Θ is compact;
3. $Q_0(\theta)$ is continuous;
4. $\hat{Q}_n(\theta)$ converges uniformly in probability to $Q_0(\theta)$,

then $\hat{\theta} \xrightarrow{P} \theta_0$.

Lemma 2.4. If the data are i.i.d., Θ is compact, $a(z_i, \theta)$ is continuous at each $\theta \in \Theta$ with probability one, and there is $d(z)$ with $\|a(z, \theta)\| \leq d(z)$ for all $\theta \in \Theta$ and $\mathbb{E}[d(z)] < \infty$, then $\mathbb{E}[a(z, \theta)]$ is continuous and $\sup_{\theta \in \Theta} \left\| n^{-1} \sum_{i=1}^n a(z_i, \theta) - \mathbb{E}[a(z, \theta)] \right\| \xrightarrow{P} 0$.

Theorem 2.7. If there is a function $Q_0(\theta)$ such that

1. $Q_0(\theta)$ is uniquely maximized at θ_0 ;
2. θ_0 is an element of the interior of a convex set Θ and $\hat{Q}_n(\theta)$ is concave; and
3. $\hat{Q}_n(\theta) \xrightarrow{P} Q_0(\theta)$ for all $\theta \in \Theta$,

then $\hat{\theta}_n$ exists with probability approaching one and $\hat{\theta}_n \xrightarrow{P} \theta_0$.

Theorem 3.1. Suppose that $\hat{\theta}$ satisfies eq. (1.1), $\hat{\theta} \xrightarrow{P} \theta_0$, and

1. $\theta_0 \in \text{interior}(\Theta)$;
2. $\hat{Q}_n(\theta)$ is twice continuously differentiable in a neighborhood $\mathcal{N}$ of θ_0 ;

3. $\sqrt{n}\nabla_{\theta}\hat{Q}_n(\theta_0) \xrightarrow{d} N(0, \Sigma)$;
4. there is $H(\theta)$ that is continuous at θ_0 and $\sup_{\theta \in \mathcal{N}} \|\nabla_{\theta\theta}\hat{Q}_n(\theta) - H(\theta)\| \xrightarrow{p} 0$;
5. $H = H(\theta_0)$ is nonsingular.

Then $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, H^{-1}\Sigma H^{-1})$.

Theorem 4.1. If the hypotheses of Theorem 3.1 are satisfied, $\hat{H} = \nabla_{\theta\theta}\hat{Q}_n(\hat{\theta})$, and $\hat{\Sigma} \xrightarrow{p} \Sigma$, then $\hat{H}^{-1}\hat{\Sigma}\hat{H}^{-1} \xrightarrow{p} H^{-1}\Sigma H^{-1}$.

Lemma 4.3. If z_i is i.i.d., $a(z, \theta)$ is continuous at θ_0 with probability one, and there is a neighborhood $\mathcal{N}$ of θ_0 such that $\mathbb{E}[\sup_{\theta \in \mathcal{N}} |a(z_i, \theta)|] < \infty$, then for any $\tilde{\theta} \xrightarrow{p} \theta_0$, $n^{-1} \sum_{i=1}^n a(z_i, \tilde{\theta}) \xrightarrow{p} E[a(z, \theta_0)]$.